

Trigonometric functions

Introduction: Prior knowledge of trigonometry

Draw a right-angled triangle on the board and mark one angle α . Students in groups, they discuss questions

Ask the students:

- What do you already know about sine and cosine?
- Where have you seen these functions before?
- What is special about the triangle we need?
- Which side is the hypotenuse?
- Which sides are the opposite and adjacent sides with respect to the angle α ?

1. Definitions

Write definitions on the board :

$$\sin(\alpha) = \frac{\text{opposite side}}{\text{hypotenuse}} \quad (1)$$

$$\cos(\alpha) = \frac{\text{adjacent side}}{\text{hypotenuse}} \quad (2)$$

2. Similar Triangles

Hand out **Exercise 1** or draw different triangles, some of them are similar.

Let them compare triangles that have the same angle but different sizes.

Discussion:

- What is different between the triangles?
- What stays the same?
- What happens to the side lengths when we make the triangle bigger?
- What happens to the ratios of the side lengths?
- What do you notice about $\sin(\alpha)$ and $\cos(\alpha)$?

The students should discover that similar triangles have different side lengths, but the ratios of the corresponding sides remain the same.

Therefore,

For a fixed angle α , the values of $\sin(\alpha)$ and $\cos(\alpha)$ are always the same.
They depend on the **angle**, not on the size of the triangle.

Ask:

“So if I tell you only the angle α , do you need to know which particular triangle I am talking about in order to know $\sin(\alpha)$?”

This motivates thinking of sine and cosine as **functions of an angle**:

$$\alpha \longmapsto \sin(\alpha), \quad \alpha \longmapsto \cos(\alpha). \quad (3)$$

3. A Useful Relationship

Continue with **Exercise 2**.

The students use Pythagoras’ theorem to determine the missing side and calculate $\sin(\alpha)$ and $\cos(\alpha)$.

Then ask them to calculate

$$\sin^2(\alpha) + \cos^2(\alpha). \quad (4)$$

Ask:

“Do you think the result is just a coincidence?”

Let the students try to explain their observation using the triangle.

If the opposite side is a , the adjacent side is b and the hypotenuse is c , then

$$\sin^2(\alpha) + \cos^2(\alpha) = \frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{a^2 + b^2}{c^2}. \quad (5)$$

Using Pythagoras’ theorem,

$$a^2 + b^2 = c^2, \quad (6)$$

we obtain

$$\boxed{\sin^2(\alpha) + \cos^2(\alpha) = 1}. \quad (7)$$

this correlation is not coincidence but holds for all angles

4. Short Application

Use **Exercise 3** as a short application.

A ladder of length 5 m leans against a wall at an angle of 60° .

Ask:

- Which side of the triangle do we know?
- Which side do we want to determine?
- Should we use sine or cosine?
- Can you write down an equation for the height h ?

The students should obtain

$$\sin(60^\circ) = \frac{h}{5 \text{ m}} \quad (8)$$

and therefore

$$h = 5 \text{ m} \cdot \sin(60^\circ). \quad (9)$$

5. Transition to the Unit Circle

Finish this part with a question that exposes a limitation of the triangle definition.

Write on the board:

$$\sin(30^\circ) \quad (10)$$

and ask:

“Can we understand this using a right-angled triangle?”

Then write:

$$\sin(120^\circ). \quad (11)$$

Ask:

“What about this one? Can a right-angled triangle have an angle of 120° ?”

The answer is no.

Nevertheless, a calculator gives a perfectly well-defined value for $\sin(120^\circ)$ and even solution for angles $\alpha > 360^\circ$.

Guiding question:

How can we extend our idea of sine and cosine so that they work for **any angle**, not only angles inside a right-angled triangle?

If no one has an answer, name **unit circle**.

* The Unit circle

“Instead of putting our angle inside a triangle, we can now think of the angle as a rotation in a coordinate system.”

- Draw circle with radius 1, a straight line through the origin and the circle in the first quadrant
- endpoint coordinates P(x,y)
- for the first quadrant: draw corresponding right-angled triangle

Questions:

- What is the length of the hypotenuse?
- What is the adjacent side?
- What is the opposite side?
- calculate sine and cosine of the angle

Since radius of the circle is 1

$$\cos(\alpha) = \frac{x}{1} = x \quad (12)$$

and

$$\sin(\alpha) = \frac{y}{1} = y. \quad (13)$$

Therefore,

$$\boxed{P = (\cos \alpha, \sin \alpha)}. \quad (14)$$

- not a completely new definition
- for angles $< 90^\circ$ definition equals our former definition with a right triangle

Going Beyond 90°

Now rotate the radius further around the circle.

Consider, for example,

$$\alpha = 120^\circ. \quad (15)$$

Ask:

- Where is the point on the circle?
- Is its x -coordinate positive or negative?
- Is its y -coordinate positive or negative?
- What does this tell us about $\cos(120^\circ)$?
- What does this tell us about $\sin(120^\circ)$?

For the second quadrant we notice:

$$\cos(\alpha) < 0, \quad \sin(\alpha) > 0. \quad (16)$$

Continue briefly through the other quadrants and let the students predict the signs of sine and cosine.

Quadrant	$\cos(\alpha)$	$\sin(\alpha)$
I	+	+
II	−	+
III	−	−
IV	+	−

Degrees and Radians

There is another way of measuring angles called radians

An intuitive way to think about radians Imagine a circle with radius 1. Start at the point $(1, 0)$ and move along the edge of the circle.

Instead of measuring how far you have turned in degrees, we can measure the angle by asking:

“How far have I travelled along the circle until I reach the point associated with the angle?”

For a circle with radius 1, this distance is exactly the angle measured in **radians**.
The circumference of the unit circle is

$$C = 2\pi. \quad (17)$$

Therefore, after travelling once around the entire circle, we have travelled a distance of 2π . This corresponds to a full rotation:

$$360^\circ = 2\pi. \quad (18)$$

Can you imagine what the following angles are in radians:

- $180^\circ = \pi$
- $90^\circ = \frac{\pi}{2}$
- $270^\circ = \frac{3\pi}{2}$,
- $360^\circ = 2\pi$.

evtl. **Exercise 4**

5. Exploring the Unit Circle

evtl. **Exercise 5.**

Let the students label the four points on the unit circle and complete the table for

$$0, \quad \frac{\pi}{2}, \quad \pi, \quad \frac{3\pi}{2}, \quad 2\pi. \quad (19)$$

Do not give them the sine and cosine values immediately.
Instead, repeatedly ask:

“What are the coordinates of the point?”

and then:

“If the coordinates are $(\cos \alpha, \sin \alpha)$, what are $\cos \alpha$ and $\sin \alpha$?”

The completed table is

α	$\cos(\alpha)$	$\sin(\alpha)$
0	1	0
$\frac{\pi}{2}$	0	1
π	-1	0
$\frac{3\pi}{2}$	0	-1
2π	1	0

Preparing the Next Discovery

Finally, ask the students to imagine that the point moves continuously around the unit circle.

“While the point moves around the circle, what happens to its y -coordinate?”

Collect a few observations:

$$0 \rightarrow 1 \rightarrow 0 \rightarrow -1 \rightarrow 0. \quad (20)$$

Discuss the following questions in groups:

“What would happen if we plotted this y -coordinate against the angle α ?”

1 Trigonometric function

So far, we have considered individual angles and determined their sine and cosine using the unit circle. Now we want to change our perspective.

“What happens to sine and cosine if we do not look at just one angle, but continuously change the angle?”

1.1 Turning values into graphs

$$\begin{aligned} \alpha = 0 &\Rightarrow \sin(\alpha) = 0, \\ \alpha = \frac{\pi}{2} &\Rightarrow \sin(\alpha) = 1, \\ \alpha = \pi &\Rightarrow \sin(\alpha) = 0, \\ \alpha = \frac{3\pi}{2} &\Rightarrow \sin(\alpha) = -1, \\ \alpha = 2\pi &\Rightarrow \sin(\alpha) = 0. \end{aligned}$$

Ask:

“What happens between these points?”

The point on the circle moves continuously, so its vertical coordinate also changes continuously.

Collect ideas, finally show simulation

Simulation https://www.geogebra.org/m/jfirtwrw?utm_source=chatgpt.com

https://animg.app/en/animation/unit-circle-generates-sine-wave-hRwsY8Hw?utm_source=chatgpt.com

The sine function describes the vertical coordinate of a point moving around the unit circle.

What Do We Notice About the Sine Function?

Look at the resulting graph together.

Discussion:

- What is the largest value the sine can have?
- What is the smallest value?

- At which angles is $\sin(\alpha) = 0$?
- Can we continue rotating the point after one complete turn?
- Can we also rotate the point in the opposite direction?
- What happens after one complete turn around the circle?

Domain Ask:

“Which angles can we put into the sine function?”

Using the unit circle, we can rotate the point by any angle. We can continue for more than one full rotation, and we can also rotate in the opposite direction, corresponding to negative angles.

Therefore, the sine function is defined for all real numbers:

$$\boxed{\text{Domain: } \alpha \in \mathbb{R}} \quad (21)$$

Range Now ask:

“Which values can come out of the sine function?”

Since $\sin(\alpha)$ is the vertical coordinate of a point on the unit circle, it can never be larger than 1 or smaller than -1 .

Therefore,

$$\boxed{\text{Range: } -1 \leq \sin(\alpha) \leq 1} \quad (22)$$

or, using interval notation,

$$\boxed{\text{Range: } [-1, 1]}. \quad (23)$$

Periodicity Finally, consider what happens after one complete rotation.

After an angle of 2π , the point has returned to exactly the same position on the unit circle. Therefore, all sine values start repeating:

$$\boxed{\sin(\alpha + 2\pi) = \sin(\alpha)}. \quad (24)$$

A function whose values repeat after a certain interval is called a **periodic function**.

The sine function has the period

$$\boxed{T = 2\pi}. \quad (25)$$

Summary From the unit circle and the graph, we have discovered for the function $f(\alpha) = \sin(\alpha)$:

- **Domain:** $\mathbb{R}$
- **Range:** $[-1, 1]$
- **Maximum:** 1
- **Minimum:** -1
- **Period:** 2π

What About Cosine?

Now return to the unit circle.

Ask:

“For the sine function, we followed the vertical coordinate. What would we have to follow to obtain the cosine function?”

Since

$$\cos(\alpha) = x_P, \quad (26)$$

the cosine describes the **horizontal position** of the point on the unit circle.

Again consider the important angles:

α	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\cos(\alpha)$	1	0	-1	0	1

Ask the students to predict the shape of the cosine graph before showing or drawing it. Then draw the sine and cosine functions in the same coordinate system.

Ask:

- What do sine and cosine have in common?
- What is different?
- Do they have the same maximum and minimum values?
- Do they have the same period?
- Could we move one graph horizontally so that it lies exactly on top of the other?

The students should notice that both functions have the same shape but start at different points.

The cosine function starts at its maximum,

$$\cos(0) = 1, \quad (27)$$

while the sine function starts at a zero crossing,

$$\sin(0) = 0. \quad (28)$$

The two graphs are shifted relative to each other by a quarter of a complete period:

$$\boxed{\cos(\alpha) = \sin\left(\alpha + \frac{\pi}{2}\right)}. \quad (29)$$

This horizontal displacement is called a **phase shift**.

Transforming Sine and Cosine Functions

So far, we have considered the basic sine and cosine functions

$$f(x) = \sin(x) \quad \text{and} \quad f(x) = \cos(x). \quad (30)$$

Both oscillate between -1 and 1 and have a period of 2π .

But real oscillations can look very different.

Discussion:

- Do all oscillations have the same maximum displacement?
- Do all oscillations take the same amount of time for one cycle?
- Do all oscillations start at the same point?
- How could we modify our sine and cosine functions to describe these different situations?

We will investigate this by changing the functions step by step. -

1. Changing the Amplitude

Start with

$$f(x) = \sin(x). \quad (31)$$

Compare it with

$$f(x) = 2 \sin(x) \quad (32)$$

and

$$f(x) = \frac{1}{2} \sin(x). \quad (33)$$

Plot the three functions in the same coordinate system.

Discussion:

- What changes?
- What stays the same?
- What are the maximum and minimum values of each function?
- Does the period change?

Let the students complete **Exercise 13**.

The students should discover that the number in front of the sine changes how far the graph moves away from its middle line.

In general,

$$f(x) = a \sin(x). \quad (34)$$

The amplitude is

$$\boxed{A = |a|}. \quad (35)$$

For example,

$$3 \sin(x)$$

oscillates between -3 and 3 .

If $a < 0$, the graph is additionally reflected across the x -axis.

2. Moving the Whole Oscillation Up and Down

Ask: How can you modify the function such that the whole graph is shifted up/down

$$f(x) = \sin(x), \quad (36)$$

$$f(x) = \sin(x) + 2, \quad (37)$$

and

$$f(x) = \sin(x) - 1. \quad (38)$$

Ask:

- Does the shape of the graph change?
- Does the amplitude change?
- Does the period change?
- What happens to the middle line of the oscillation?

The students should discover that adding a number moves the entire graph vertically. In general,

$$f(x) = \sin(x) + d. \quad (39)$$

The middle line is now

$$\boxed{y = d}. \quad (40)$$

Therefore,

- $d > 0$: the graph moves upwards,
- $d < 0$: the graph moves downwards.

3. Changing the Period

Return to the basic sine function and compare

$$f(x) = \sin(x) \quad (41)$$

with

$$f(x) = \sin(2x). \quad (42)$$

Let the students first look at the graphs without giving them the rule.

Discussion:

- How many complete oscillations does $\sin(x)$ make between 0 and 2π ?
- How many complete oscillations does $\sin(2x)$ make?
- Which function repeats faster?
- What is the period of $\sin(2x)$?
- Does the amplitude change?

Let the students complete **Exercise 14**.

Then consider another example:

$$f(x) = \sin\left(\frac{1}{2}x\right). \quad (43)$$

Ask the students to predict what will happen before plotting the function.

For the general function

$$f(x) = \sin(bx), \quad (44)$$

the period is

$$T = \frac{2\pi}{|b|}. \quad (45)$$

Thus, a larger value of $|b|$ means a shorter period and therefore more oscillations in the same interval.

The factor outside the sine changes the **amplitude**.
The factor multiplying x inside the sine changes the **period**.

4. Shifting the Oscillation Horizontally

Now compare

$$f(x) = \sin(x) \quad (46)$$

and

$$f(x) = \sin\left(x - \frac{\pi}{2}\right). \quad (47)$$

Ask:

- Has the shape of the graph changed?
- Has the amplitude changed?
- Has the period changed?
- Where is the first graph at $x = 0$?
- Where is the second graph at $x = 0$?
- In which direction has the graph moved?

The students should discover that the entire oscillation has been shifted horizontally.

For

$$f(x) = \sin(x + c), \quad (48)$$

the graph is shifted by

$$-c. \quad (49)$$

Therefore,

- $\sin(x - c)$ is shifted to the **right** by c ,
- $\sin(x + c)$ is shifted to the **left** by c .

This horizontal shift is called the **phase shift**.

5. Putting Everything Together

We can now combine all four transformations.

A general transformed sine function can be written as

$$\boxed{f(x) = a \sin(bx + c) + d}. \quad (50)$$

Similarly, for cosine,

$$\boxed{f(x) = a \cos(bx + c) + d}. \quad (51)$$

The parameters have the following effects:

Parameter	What does it change?	Value
a	Amplitude	$A = a $
b	Period	$T = \frac{2\pi}{ b }$
c	Horizontal position / phase	shift = $-\frac{c}{b}$
d	Vertical position	middle line $y = d$

Important: The same rules apply to both sine and cosine.

6. Can We Read a Function?

Before calculating anything, show the students an example:

$$f(x) = 2 \sin\left(3x - \frac{\pi}{2}\right) + 1. \quad (52)$$

Ask them to identify the four parameters:

$$a = 2, \quad b = 3, \quad c = -\frac{\pi}{2}, \quad d = 1. \quad (53)$$

Then determine the properties step by step:

$$\text{Amplitude: } A = 2, \quad (54)$$

$$\text{Period: } T = \frac{2\pi}{3}, \quad (55)$$

$$\text{Phase shift: } -\frac{c}{b} = \frac{\pi}{6}, \quad (56)$$

$$\text{Middle line: } y = 1. \quad (57)$$

Therefore, the function describes an oscillation with amplitude 2 around the middle line $y = 1$, with a period of $2\pi/3$, shifted by $\pi/6$ to the right.

7. From Mathematics to Physics

Finally, return to the question from the beginning:

“What physical systems or phenomena can be described by sine and cosine functions?”

Possible answers:

- a swinging pendulum

- a mass attached to a spring
- a vibrating string
- sound waves
- water waves (approximately)
- alternating current
- electromagnetic waves, such as light